A construction of four-dimensional Yang–Mills theory

First sections for review; the detailed proofs are incomplete.

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Part I
The theorem and its proof

1 The theorem and its proof

The construction starts from probability measures on finite periodic lattices. Removing the lattice spacing at fixed physical volume gives the first limit; compactness then gives an infinite-volume limit along a refinement of any prescribed increasing sequence of volumes. The resulting Euclidean correlations decompose into phases. One selected phase gives a relativistic theory, while the mixture and every phase admit the same finite operator product expansions. A logarithmic correction in a degree-four two-point function excludes a specified class of free-field profiles.

We first define the objects and the exact conclusions of the theorem. The five propositions at the end of this section describe the construction and explain how those conclusions fit together. The mathematical statements follow the formal source at the fixed revision recorded in [1]. Their correspondence with individual declarations is recorded separately; the detailed proofs in Part II remain to be written.

1.1 The lattice family

Throughout, $\mathbb{N} = \{0, 1, \ldots\}$ and the fourth coordinate of $\mathbb{R}^4$ is time. A gauge group consists of an integer $m \geq 1$, a closed subgroup $G \subset U(m)$ and a real linear subspace $\mathfrak{g} \subset M_m(\mathbb{C})$ satisfying

$$X \in \mathfrak{g} \quad \Longleftrightarrow \quad \exp(sX) \in G \text{ for every } s \in \mathbb{R}\,. \tag{1.1}$$

We call $\mathfrak{g}$ simple if some $X, Y \in \mathfrak{g}$ have $[X, Y] \neq 0$ and every real subspace $J \subset \mathfrak{g}$ with $[\mathfrak{g}, J] \subset J$ equals 0 or $\mathfrak{g}$. The unitary realization is part of the data; connectedness of G is not required.

For a positive integer ℓ, let $\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$. A link configuration assigns $U(y, \mu) \in G$ to each $y \in \Lambda_\ell$ and $\mu \in \{1, 2, 3, 4\}$. The plaquette and its Wilson energy are

$$\begin{aligned} U_{y;\mu\nu} &= U(y, \mu)U(y + e_\mu, \nu)U(y + e_\nu, \mu)^{-1}U(y, \nu)^{-1}, \\ E_\ell(U) &= \sum_{y \in \Lambda_\ell} \sum_{\mu < \nu} \left(1 - \frac{1}{m} \operatorname{Re} \operatorname{tr} U_{y;\mu\nu}\right). \end{aligned} \tag{1.2}$$

For normalized Haar measure h on G and $\beta \in \mathbb{R}$, define

$$\mu_{\ell,\beta}(dU) = \frac{e^{-\beta E_\ell(U)}}{Z_{\ell,\beta}} \prod_{y,\mu} h(dU(y, \mu)), \qquad Z_{\ell,\beta} = \int e^{-\beta E_\ell(U)} \prod_{y,\mu} h(dU(y, \mu))\,. \tag{1.3}$$

Since each plaquette summand lies in $[0, 2]$, we have $0 \leq E_\ell \leq 12\ell^4$; hence the normalizer is finite and strictly positive. A gauge transformation $\gamma : \Lambda_\ell \to G$ acts by

$$U^\gamma(y, \mu) = \gamma(y)U(y, \mu)\gamma(y + e_\mu)^{-1}\,.$$

Definition 1.1 (Regular Wilson family)**.** A Wilson family $\mathcal{W}$ consists of strictly increasing sequences L_p, D_N of positive integers, real couplings $\beta_{N,p}$, normalized Haar measure h, a set $\mathcal{T}$ of tokens and complex insertions $I_{N,p}(v, y; U)$ for every token v, site $y \in \Lambda_{L_pD_N}$ and link configuration U on that lattice. Put $a_N = D_N^{-1}$ and $\mu_{N,p} = \mu_{L_pD_N,\beta_{N,p}}$. The family is regular if the following conditions hold.

Reg1. There exists $b_* > 0$ such that, for every $\varepsilon > 0$, there exists N_0 for which

$$\left| \frac{\beta_{N,p}}{\log(1/a_N)} - b_* \right| < \varepsilon \qquad (N \geq N_0,\ p \in \mathbb{N}) \,. \tag{1.4}$$

Reg2. Every insertion is continuous in U.

Reg3. Every insertion is invariant under the gauge transformations above.

Reg4. For each $v \in \mathcal{T}$ there exists a finite stencil $\Sigma_v \subset \mathbb{Z}^4$ such that, for every N, p, y, the insertion depends only on links with both endpoints in $y + (\Sigma_v \bmod L_p D_N)$.

The fixed stencil shrinks to a point in physical units. The insertion may depend on both cutoff and volume. Regularity specifies neither a classical polynomial represented by a token nor a value for the positive slope b_*. The construction will make these choices; they are not additional requirements in this definition.

Write $r_\ell(j)$ for the representative of $j \in \mathbb{Z}/\ell\mathbb{Z}$ in $(-\ell/2, \ell/2]$. The physical site is $\hat{y} = a_N (r_{L_p D_N}(y^\mu))_{\mu=1}^4$. For a token list $v \in \mathcal{T}^n$ define

$$\begin{aligned} M_{N,p}(v; y) &= \int \prod_{i=1}^{n} I_{N,p}(v_i, y_i; U)\, \mu_{N,p}(dU), \\ C_{N,p}(v; f) &= a_N^{4n} \sum_{y \in \Lambda_{L_p D_N}^n} f(\hat{y}_1, \ldots, \hat{y}_n) M_{N,p}(v; y) \,. \end{aligned} \tag{1.5}$$

The separated configuration space is

$$\mathrm{Conf}_n = \{x \in (\mathbb{R}^4)^n : x_i \neq x_j \text{ whenever } i \neq j\} \,. \tag{1.6}$$

A separated test f is a globally smooth complex function whose closed support is compact and contained in Conf_n. It is p-admissible if its support lies in $((-L_p/2, L_p/2)^4)^n$. These are joint tests of all points; they need not be products of one-point tests. Arity zero is included throughout, with empty products equal to 1.

Definition 1.2 (Ordered limit)**.** For $P : \mathbb{N} \to \mathbb{N}$ and kernels $K_n(v; x)$, the family $\mathcal{W}$ converges along P to K if, for every $n \in \mathbb{N}$, $v \in \mathcal{T}^n$ and separated test f, the test is $P(k)$-admissible eventually in k, and there exist numbers $R_p \in \mathbb{C}$ such that

$$\begin{aligned} \lim_{N \to \infty} C_{N,p}(v; f) &= R_p \qquad\qquad (f \ p\text{-admissible}), \\ \lim_{k \to \infty} R_{P(k)} &= \int_{(\mathbb{R}^4)^n} f(x) K_n(v; x)\, dx \,. \end{aligned} \tag{1.7}$$

The first limit uses the entire ultraviolet sequence at each admissible volume. The numbers R_p may be assigned arbitrarily at inadmissible volumes. A strictly increasing P makes each compact test support admissible eventually, since $L_{P(k)} \to \infty$.

1.2 The continuum objects

Write $x = (\vec{x}, t(x))$ for spacetime, with Euclidean norm $|x|$ and

$$\mathsf{m}(x) = t(x)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{x : |\vec{x}| \leq t(x)\} \,. \tag{1.8}$$

A vector is spacelike if $\mathsf{m}(x) < 0$. Let $\mathcal{S} = \mathcal{S}(\mathbb{R}^4; \mathbb{C})$, with its Schwartz topology. Hilbert inner products are antilinear in the first variable.

Definition 1.3 (Relativistic data)**.** A relativistic datum consists of a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ of labels, a linear subspace $\mathcal{D} \subset \mathcal{H}$, a vector $\Omega \in \mathcal{D}$ and fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$ linear over $\mathbb{C}$ in $\ell \in \mathcal{L}$ and $f \in \mathcal{S}$. It includes an antilinear map $\ell \mapsto \ell^*$, a strongly continuous unitary translation representation $T(b)$ and a projection-valued Borel measure E on $\mathbb{R}^4$. Its diagonal finite measures satisfy $\mu_\psi(A) = \langle \psi, E(A)\psi \rangle$ on Borel sets.

There is also a strongly continuous unitary representation U of a topological group $\mathcal{P}$, identified by a homeomorphism κ with the proper orthochronous Poincaré group. Writing $\kappa(g) = (\Lambda_g, b_g)$, this identification respects identity and the multiplication

$$(\Lambda, b)(\Lambda', b') = (\Lambda\Lambda', \Lambda b' + b)\,. \tag{1.9}$$

Here Λ is invertible, has determinant 1, preserves m and maps $\overline{V}_+$ onto itself. Pure translations $\tau(b) = \kappa^{-1}(1, b)$ satisfy $U(\tau(b)) = T(b)$. For each $g \in \mathcal{P}$ there is a complex linear bijection $\rho(g)$ of $\mathcal{L}$. The action on tests is $(g \cdot f)(x) = f(\Lambda_g^{-1}(x - b_g))$.

A projection-valued Borel measure here is normalized by $E(\mathbb{R}^4) = 1$, has $E(\varnothing) = 0$, satisfies $E(A)E(B) = E(A \cap B)$ and $E(A)^* = E(A)$ on Borel sets, and is countably additive in the strong operator topology. One can assign arbitrary operators to non-Borel sets, which none of the assertions uses. The label map * is not required to be an involution. No continuity or finite-dimensionality condition on ρ is included in the data.

Definition 1.4 (Wightman conditions)**.** The relativistic datum satisfies the following conditions, with $p \cdot b$ the Euclidean pairing.

W0. For every $\psi \in \mathcal{H}$ and $b \in \mathbb{R}^4$,

$$\langle \psi, T(b)\psi \rangle = \int e^{ip\cdot b}\, \mu_\psi(dp), \qquad E(\mathbb{R}^4 \setminus \overline{V}_+) = 0\,. \tag{1.10}$$

Every vector fixed by all $T(b)$ belongs to $\mathbb{C}\Omega$.

W1. The vacuum has norm one, $\mathcal{D}$ is dense, and

$$\mathcal{D} = \operatorname{span}_{\mathbb{C}}\{\Phi(\ell_1, f_1) \cdots \Phi(\ell_j, f_j)\Omega : j \in \mathbb{N},\ \ell_i \in \mathcal{L},\ f_i \in \mathcal{S}\}\,. \tag{1.11}$$

For every $\ell \in \mathcal{L}$ and $\psi \in \mathcal{D}$, the map $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ into $\mathcal{H}$ in norm. For every ℓ, f and $\psi, \chi \in \mathcal{D}$,

$$\langle \Phi(\ell, f)\psi, \chi \rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi \rangle\,. \tag{1.12}$$

W2. The maps ρ form a representation: $\rho(1) = 1$ and $\rho(gg') = \rho(g)\rho(g')$. For every $g \in \mathcal{P}$, $U(g)\mathcal{D} \subset \mathcal{D}$ and $U(g)\Omega = \Omega$. For every g, ℓ, f and $\psi \in \mathcal{D}$,

$$U(g)\Phi(\ell, f)\psi = \Phi(\rho(g)\ell, g \cdot f)U(g)\psi\,. \tag{1.13}$$

W3. If $\mathsf{m}(x - y) < 0$ for every $x \in \operatorname{supp} f$ and $y \in \operatorname{supp} f'$, then, for every $\ell, \ell' \in \mathcal{L}$ and $\psi \in \mathcal{D}$,

$$\Phi(\ell, f)\Phi(\ell', f')\psi = \Phi(\ell', f')\Phi(\ell, f)\psi\,. \tag{1.14}$$

The word span is the domain itself, and includes the vacuum as the empty word. The symmetry group is the Poincaré group itself rather than its double cover. The sign convention in (1.10) is converted to the usual Minkowski pairing by reversing spatial momentum; the forward cone is unchanged.

Definition 1.5 (Hierarchy)**.** A hierarchy over $\mathcal{W}$ contains the following data.

H1. A relativistic datum of Definition 1.3.

H2. For every $d \in \mathbb{N}$, a finite-dimensional complex normed space F_d and a complex linear map $\iota_d : F_d \to \mathcal{L}$. Set $\mathbb{F} = \bigsqcup_{d\in\mathbb{N}} F_d$, $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$.

H3. A projection $\pi : \mathcal{T} \to \mathbb{F}$.

H4. A compact metric space X with its Borel measure ν and a selected point $\xi_* \in X$.

H5. For every $n \in \mathbb{N}$ and $s \in X \sqcup \{\mathrm{mix}\}$, a correlation function $S^s(a; x) \in \mathbb{C}$ on $\mathbb{F}^n \times (\mathbb{R}^4)^n$.

H6. For every $n \in \mathbb{N}$, a continuation $\mathsf{F}_n(a; z, \vec{x}) \in \mathbb{C}$ on $\mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n$.

H7. For every $j \in \mathbb{N}$, a finite channel set Ch_j and fields $O_{j,\kappa} \in F_j$. For every finite index set I, $a \in \mathbb{F}^I$ and $\kappa \in \mathrm{Ch}_j$, there is a coefficient function $c_{j,\kappa}(a; x, o)$ on $(\mathbb{R}^4)^I \times \mathbb{R}^4$.

H8. An element $\Theta \in F_4$.

The degree is an index used in the product estimates below. The definition imposes no inclusions between different F_d, no prescribed curvature interpretation and no compatibility of the ι_d across degrees. The functions S^s are defined even on collision diagonals, but the conclusions constrain them only at separated points.

1.3 Reconstruction and phases

The mixture will be the limit of lattice moments. The point ξ_* instead selects the Euclidean correlations that continue to the relativistic fields. These two roles are related by a phase average, not by identifying the two states.

Definition 1.6 (Reconstruction properties)**.** A hierarchy has the reconstruction properties if the following hold.

R1. Its relativistic datum satisfies W0–W3.

R2. For every nonzero $\ell \in \mathcal{L}$, there exists $f \in \mathcal{S}$ with $\Phi(\ell, f) \neq 0$.

R3. Each ι_d is injective.

R4. Some $a \in \mathbb{F}$ and $f \in \mathcal{S}$ satisfy $\Phi(\iota(a), f) \neq 0$.

R5. The map $a \mapsto \iota(a)$ from $\mathbb{F}$ onto $\mathcal{L}$ is surjective.

R6. The map $\pi : \mathcal{T} \to \mathbb{F}$ is surjective.

R7. The measure ν has total mass one.

R8. Every nonempty open subset of X has positive ν-measure.

R9. For every state s, arity n, list $a \in \mathbb{F}^n$, $x \in \mathrm{Conf}_n$ and permutation σ,

$$S^s((a_{\sigma(i)})_i; (x_{\sigma(i)})_i) = S^s(a; x)\,. \tag{1.15}$$

R10. For every s, n, a, the function $S^s(a; \cdot)$ is smooth on Conf_n.

R11. For every n, a and $x \in \mathrm{Conf}_n$, the function $\xi \mapsto S^\xi(a; x)$ is continuous.

R12. For every n, a and $x \in \mathrm{Conf}_n$,

$$\int_X S^{\xi}(a;x)\,\nu(d\xi) = S^{\mathrm{mix}}(a;x)\,. \tag{1.16}$$

R13. For every n, a and fixed real spatial configuration $\vec{x}$, the function $z \mapsto \mathsf{F}_n(a;z,\vec{x})$ is holomorphic on

$$\mathcal{Z}_n = \{z \in \mathbb{C}^n : \operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n\}\,. \tag{1.17}$$

R14. For every n, a and real configuration x with $t(x_1) < \cdots < t(x_n)$,

$$\mathsf{F}_n(a;(it(x_i))_i,(\vec{x}_i)_i) = S^{\xi_*}(a;x)\,. \tag{1.18}$$

R15. For every n, a, real heights $h_1 < \cdots < h_n$ and tests $f_1, \dots, f_n \in \mathcal{S}$,

$$\lim_{r\downarrow 0} \int_{(\mathbb{R}^4)^n} \mathsf{F}_n(a;(t(x_i)+irh_i)_i,(\vec{x}_i)_i) \prod_{i=1}^n f_i(x_i)\,dx = \langle \Omega, \Phi(\iota(a_1),f_1)\cdots\Phi(\iota(a_n),f_n)\Omega\rangle\,. \tag{1.19}$$

In (1.19) we retain the formal source's convention that the integral of a nonintegrable function is zero. No separate positive-height integrability assertion is part of R15. The integrals in the lattice limit and (1.16) are ordinary absolutely convergent integrals, by compact support and smoothness in the first case and by phase continuity and compactness in the second.

1.4 Products and the trace test

The product expansion replaces a cluster by finitely many single fields at its center. One choice of coefficients must work with every list of distant spectators and in every state. The threshold on the cluster radius, however, may depend on the fixed spectator fields.

For finite index sets I, J, write $S^s(a,x;b,y)$ for the correlation on the concatenated lists, enumerated in one fixed order on $I \sqcup J$. For $r > 0$, $M \geq 0$ and $\eta > 0$, call (x,o) an (r,M,η)-cluster if

$$|x_i - o| \leq Mr \quad (i \in I), \qquad |x_i - x_{i'}| \geq \eta r \quad (i \neq i')\,. \tag{1.20}$$

For $c > 0$, call y c-clear of o if

$$|o - y_j| \geq c \quad (j \in J), \qquad |y_j - y_{j'}| \geq c \quad (j \neq j')\,. \tag{1.21}$$

Set $w(a) = \sum_{i\in I} \deg a_i$.

Definition 1.7 (Operator product expansion)**.** The hierarchy has the OPE property if both assertions below hold.

O1. For every finite I, $a \in \mathbb{F}^I$, $M \geq 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\kappa \in \mathrm{Ch}_j$, there exists $\delta > 0$ such that, for every $0 < r < \delta$ and every (r,M,η)-cluster,

$$|c_{j,\kappa}(a;x,o)| \leq r^{j-w(a)-\varepsilon}\,. \tag{1.22}$$

O2. For every finite I, J, $a \in \mathbb{F}^I$, $b \in \mathbb{F}^J$, $M \geq 0$, $\eta, c, \varepsilon > 0$ and $D \in \mathbb{N}$, there exists $\delta > 0$ such that, for every $0 < r < \delta$, every (r, M, η)-cluster, every c-clear spectator configuration and every $s \in X \sqcup \{\mathrm{mix}\}$,

$$\left| S^s(a, x; b, y) - \sum_{j=0}^{D} \sum_{\kappa \in \mathrm{Ch}_j} c_{j,\kappa}(a; x, o) S^s((j, O_{j,\kappa}), o; b, y) \right| \leq r^{D+1-w(a)-\varepsilon} . \tag{1.23}$$

The coefficient functions are those already chosen in H7. They do not depend on spectators, state or truncation degree. Both estimates have constant 1, with the dependence absorbed into δ. For small enough r, $Mr < c$ ensures that all points in the tested correlations are separated. These are finite-truncation estimates; they assert neither an infinite convergent expansion nor uniqueness of its coefficients.

The last conclusion concerns a single two-point function in the selected phase:

$$G_\Theta(r) = S^{\xi_*}((4, \Theta), (4, \Theta); -re_4/2, re_4/2) . \tag{1.24}$$

To specify the comparison class, a mode consists of $\mathsf{m} \geq 0$, integers $j, q \in \mathbb{N}$ and a continuous real function h on $\mathbb{R}^3$ satisfying $h(\lambda p) = \lambda^j h(p)$ for every $\lambda > 0$. Set

$$E_{\mathsf{m}}(p) = \sqrt{\mathsf{m}^2 + |p|^2}, \qquad K_{\mathsf{m},q,h}(r) = \int_{\mathbb{R}^3} \frac{E_{\mathsf{m}}(p)^q h(p)}{2E_{\mathsf{m}}(p)} e^{-rE_{\mathsf{m}}(p)} \, dp . \tag{1.25}$$

The integrand can be assigned value zero at a zero denominator, a set of measure zero. A finite Wick profile is

$$W(r) = \sum_{\alpha=1}^{A} c_\alpha \prod_{\beta=1}^{n_\alpha} K_{\mathsf{m}_{\alpha\beta}, q_{\alpha\beta}, h_{\alpha\beta}}(r), \qquad c_\alpha \in \mathbb{C}, \quad A, n_\alpha \in \mathbb{N} . \tag{1.26}$$

Empty products are allowed.

Definition 1.8 (Trace test)**.** The hierarchy passes the trace test if the following hold.

T1. For every $f \in \mathcal{S}$, $\langle \Omega, \Phi(\iota_4 \Theta, f)\Omega \rangle = 0$.

T2. The number $A_G = 3 \dim_{\mathbb{R}} \mathfrak{g}/\pi^4$ is positive.

T3. The following limit holds in $\mathbb{C}$ over all positive radii:

$$\lim_{r \downarrow 0} r^8 \big(\log(1/r)\big)^2 G_\Theta(r) = \frac{3 \dim_{\mathbb{R}} \mathfrak{g}}{\pi^4} . \tag{1.27}$$

T4. For every finite Wick profile W and every $\delta > 0$, there exists $r \in (0, \delta)$ with $G_\Theta(r) \neq W(r)$.

The public trace test concerns the chosen degree-four field and the time axis. Its definition alone supplies no stress tensor. The construction identifies this field with a centered stress trace, as used in the proof of Proposition 1.13 below.

1.5 The main theorem

A hierarchy over $\mathcal{W}$ has the main properties along P if:

P1. $\mathcal{W}$ converges along P in Definition 1.2 to $K_n(v;x) = S^{\mathrm{mix}}(\pi(v_1),\dots,\pi(v_n);x)$;

P2. it has properties R1–R15;

P3. it has properties O1–O2;

P4. it has properties T1–T4.

These four assertions refer to one hierarchy, including the very same spectator fields in P1 and in the OPE.

Theorem A. *For every gauge group $(m,G,\mathfrak{g})$ with $\mathfrak{g}$ simple, there exists a regular Wilson family $\mathcal{W}$ such that, for every strictly increasing $p:\mathbb{N}\to\mathbb{N}$, there exist a strictly increasing $q:\mathbb{N}\to\mathbb{N}$ and a hierarchy over $\mathcal{W}$ with the main properties along $p\circ q$.*

The lattice data are chosen before p. The refinement, field spaces, phases, selected relativistic theory, product coefficients and trace field may depend on p. The same refinement works for all arities, tokens and separated tests. The lattice limit is the mixture; the relativistic theory is attached to ξ_*.

The theorem does not assert uniqueness of the infinite-volume limit, universality with respect to lattice data, or convergence along simultaneous cutoff and volume limits. It does not assert a mass gap, clustering of the mixture, agreement of phases, or that an individual phase is a lattice-state limit. It gives no contact extension across collisions, and excludes only the finite profiles in (1.26), not all composites of Gaussian fields or continuous-mass generalized free fields. No Euclidean covariance or reflection-positivity assertion is included among P1–P4; those properties are established internally in the construction.

1.6 The principal propositions

The following propositions assert one joint choice of objects. For each gauge group there exists one family satisfying Proposition 1.9; for each increasing p there exist one increasing q and one hierarchy satisfying Propositions 1.10–1.13 together. The latter assertions concern this chosen family and hierarchy. They are not assertions about an arbitrary regular family or about an arbitrary table of smooth correlations. This joint choice is essential: separate existence of objects satisfying the separate conclusions would not imply Theorem A.

Removing the ultraviolet cutoff. The short-distance integration must control the observables as well as the density. At fixed volume the bare coupling changes with the cutoff, so comparison at a fixed bare coupling cannot establish convergence. Instead, each cutoff trajectory is tuned to one common effective coupling at unit scale.

Proposition 1.9 (Ultraviolet construction)**.** *For every gauge group with simple $\mathfrak{g}$, there exists a regular Wilson family $\mathcal{W}$ in the joint construction with the following property. For every $p,n\in\mathbb{N}$ and $v\in\mathcal{T}^n$, there exists a function $K_{p,v}$ smooth on*

$$\{x\in(\mathbb{R}^4)^n : x_i - x_j \notin L_p\mathbb{Z}^4 \text{ whenever } i\neq j\} \tag{1.28}$$

such that, for every p-admissible separated test f,

$$\lim_{N\to\infty} C_{N,p}(v;f) = \int f(x)K_{p,v}(x)\,dx\,. \tag{1.29}$$

Idea of the proof. An exactly normalized, gauge-covariant block kernel preserves the partition integral. Gaussian integration around a stationary background and a connected polymer expansion estimate the small-field part, while an energy lower bound controls the retained large-field part. The marginal curvature-square coefficient is absorbed into the running coupling by an exact identity of densities. The choices of block ratios and entry couplings make the accumulated coupling error smaller than the total logarithmic scale. Marked insertions evolve through these same steps, with a fixed reference normalization for each finite degree. Their uniform tails allow the fixed-level limit to imply the full fixed-volume limit: first make the tail small, then compare large cutoffs at that fixed level.

Unresolved. The Part II proof must supply the nonlinear-step estimates, attach the one-loop coefficient to the actual stationary determinant, and prove convergence of the tuned fixed-level data with uniform observable tails. The stated existence and full-spacing limit have been checked at the formal assembly; these analytic proofs have not yet been written here.[1]

Taking infinite volume and resolving the mixture. Local estimates independent of the torus permit a common compactness extraction. Their reflection-positive Hilbert space may have more than one ground state. Compressing bounded slab insertions to that ground space gives the algebra whose characters describe the phases.

Proposition 1.10 (Continuum limit and phases)**.** *For the jointly chosen family of Proposition 1.9 and every strictly increasing $p : \mathbb{N} \to \mathbb{N}$, there exist a strictly increasing $q : \mathbb{N} \to \mathbb{N}$ and hierarchy data H1–H8 such that P1 and R7–R12 hold. The same hierarchy satisfies Propositions 1.11–1.13.*

Idea of the proof. Volume-uniform chessboard bounds control separated correlations. A positive transfer kernel and complex interpolation give equicontinuity, and countable extraction gives a common refinement for all finite label bases and compact sets. Lattice summation by parts and corrected higher-degree representatives identify positional derivatives. The full fixed-volume limits of Proposition 1.9 identify the inner limits, removing the auxiliary ultraviolet refinement.

For the phase decomposition, powers of a slab operator are bounded with an exterior-word factor incurred once, independently of the number of powers. Taking roots removes this factor and gives a bounded operator. Continuous rotations, proved by stress-shell identities and comparison with free Maxwell fields, allow the time and a spatial axis to be exchanged. This exchange proves commutativity of the ground-space algebra. Its compact spectrum and the measure representing the vacuum give (1.16).

Unresolved. The first limit section must state the actual ultraviolet estimates needed for compactness and for corrected derivative representatives. The later symmetry and phase sections must prove the exterior-flux estimate and commutativity before using the character decomposition; compactness alone does not give these conclusions.[2]

Continuing a selected phase. A phase with one-dimensional ground space is selected without imposing an extra condition on its trace correlations. The continuation is built from bounded slab insertions separated by positive heat evolution. Bounds as the slabs become thin control its boundary values.

[1] Formal correspondence at revision `3456e5e`: `Construction.lean`, lines 15–40; `HierarchyBridge.lean`, lines 16–26; `YM/FullVolume/Profiles.lean`, lines 25–47..

[2] Formal correspondence at revision `3456e5e`: `YM/Extraction/Limits.lean`, lines 26–43; `HierarchyBridge.lean`, lines 44–57 and 82–119; the source phase construction is `YM/Phase/Mixture.lean`..

Proposition 1.11 (Relativistic reconstruction)**.** *For the jointly chosen hierarchy in Proposition 1.10, properties R1–R6 and R13–R15 hold, with the same F_d, π, ι_d, ξ_* and S^{ξ_*}.*

Idea of the proof. Polynomial thin-slab bounds yield boundary distributions. Positivity realizes their field words on a Hilbert space; prepending a field defines the operators on the word span. Continued rotation Ward identities give Lorentz covariance, and the positive-energy half-space together with that covariance gives forward-cone spectrum. Agreement of the two analytic orderings at a spacelike adjacent pair proves local commutativity. Faithfulness follows by recovering the Euclidean correlations from a vanishing continued field. The quotient by all separated null correlations makes each ι_d injective, and compatible finite-degree fields cover the label space. The unit field and faithfulness supply a nonzero field action.

Unresolved. The reconstruction section must prove the joint boundary estimates, the spacelike analytic gluing and their application to the chosen phase. The public properties, including norm continuity of smearing and the precise product-test boundary limit, have been checked in the formal conversion; the analytic arguments are still deferred.[3]

Expanding a local cluster. The lattice evolution expresses a cluster as a local insertion plus a remainder. A finite family of separating correlations reads its coefficients degree by degree. An operator-norm remainder estimate in the commutative ground algebra then applies to every character and the mixed vacuum at once.

Proposition 1.12 (Common operator product expansion)**.** *The fields and coefficient functions $H7$ of the jointly chosen hierarchy satisfy O1–O2 of Definition 1.7. One coefficient system serves every spectator list, state and finite truncation degree.*

Idea of the proof. Integrate the clustered product from its scale toward the spectator scale, retaining its components of degree at most D. The remainder gains the next power of the cluster radius, with an arbitrarily small exponent loss absorbing the logarithmic factors. Choose the separating correlations once, before any spectator is prescribed. Apply the resulting algebra norm bound before evaluating in a phase; this gives the uniformity in s in (1.23). The permitted dependence of the radius threshold on the spectator fields is retained.

Unresolved. The product section must prove the marked cluster estimate and the passage from the chosen separating correlations to the algebra-norm remainder. The exact eventual-radius quantifiers of O1–O2 have been checked against the formal statements, but these estimates have not yet been reproduced.[4]

Detecting the logarithmic correction. The construction supplies a conserved stress tensor and centers its trace in the selected phase. Short-distance comparison with free Maxwell fields determines the trace's leading behavior along suitable scales. Positivity of its spectral measure and the ability to fit a subsequence of every sequence of radii give the full right limit.

Proposition 1.13 (Trace law and finite-profile exclusion)**.** *The field $\Theta \in F_4$ of the jointly chosen hierarchy satisfies T1–T4 of Definition 1.8 in the selected theory of Proposition 1.11.*

[3] Formal correspondence at revision `3456e5e`: `PublicTheory.lean`, lines 93–319; `HierarchyBridge.lean`, lines 82–133; boundary and locality proofs are organized in `YM/Work/C01.lean` through `C29.lean`..

[4] Formal correspondence at revision `3456e5e`: `ChallengeSpec.lean`, lines 58–80; `HierarchyBridge.lean`, lines 68–80; `YM/OPE/Orbits/Kernel.lean`, lines 62–77..

Idea of the proof. The trace comparison contributes one inverse running coupling at each insertion. Its logarithmic running therefore produces the inverse square logarithm in (1.27). The one-loop coefficient enters both the trace normalization and the coupling; their ratio cancels from the final amplitude. Thus that amplitude alone does not check the numerical beta-function coefficient.

The positive spectral Laplace representation makes G_Θ nonincreasing. Comparison at fitted separations on either side of a prescribed radius then gives the same asymptotic there. Fitting a subsequence of any proposed counterexample sequence proves the full-radius limit. Finite products and sums of the kernels in (1.25) instead have expansions with integer powers and nonnegative powers of the logarithm. The finite-order asymptotic estimates for those expansions preclude a finite nonzero limit after multiplication by $r^8 \log^2(1/r)$, which proves T4.

Unresolved. The trace section must prove the stress normalization, flexible-scale comparison and spectral sandwich, and the finite mass-shell expansion used in the exclusion. The trace amplitude and the conversion of the comparison class have been checked directly; no statement about continuous-mass generalized free fields is inferred.[5]

Deduction of Theorem A from the principal propositions. Fix the gauge group and choose the family of Proposition 1.9, including its insertions and positive regularity slope. For a prescribed strictly increasing p, take the common q and hierarchy of Proposition 1.10. This gives P1 and R7–R12 on that refinement. Proposition 1.11 supplies the remaining reconstruction properties, hence P2. Propositions 1.12 and 1.13 give P3 and P4 for those same objects. □

The joint existence required here follows in the formal proof by choosing the block-kernel penalty before the family, specializing it to 1, and using the full ultraviolet sequence with the prescribed volume sequence. The detailed construction then puts the limit table, selected theory, phases, product coefficients and trace on one refinement. This checks the logical reduction and the identities of its objects; it does not replace the deferred analytic proofs above.[6]

Part II
The construction

2 Lattice fields

The lattice measure and the local observables are fixed together, before any sequence of volumes is prescribed. This section proves the exact symmetries of the lattice law and defines the classical labels of local observables. The normalization of these polynomials has two stages: their Taylor coefficients reproduce classical fields, and their coefficients are subsequently changed by the renormalization transformation. Keeping these stages distinct will allow us to use one finite stencil at every cutoff.

Throughout this section, $G \subset U(m)$ is the closed matrix group of the theorem, with real Lie algebra $\mathfrak{g}$. The arguments here do not require simplicity or connectedness. The Frobenius norm on matrices is denoted by $\|\cdot\|_{\mathrm{F}}$.

[5] Formal correspondence at revision `3456e5e`: `TraceBridge.lean`, lines 14–44; `YM/TraceExtension/Verse/Cape.lean`, lines 53–88; `YM/TraceExtension/Wick/Log.lean`, lines 96–108..

[6] Formal correspondence at revision `3456e5e`: `Solution.lean`, lines 4–10, specializes `YangMills.constructed`; the common witnesses and all four properties are assembled in `Construction.lean`, lines 15–40..

2.1 The Wilson measure

Only positive links are independent variables. For $\ell \geq 1$, put $\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$ and assign $U(y,\mu) \in G$ to each link from y to $y + e_\mu$. Transport on the reverse link is $U(y,\mu)^{-1}$. If w is a lattice walk from x to z, write $U(w)$ for the ordered product of its link variables. A gauge transformation $\gamma : \Lambda_\ell \to G$ acts by

$$U^\gamma(y,\mu) = \gamma(y)U(y,\mu)\gamma(y+e_\mu)^{-1}, \qquad U^\gamma(w) = \gamma(x)U(w)\gamma(z)^{-1}\,. \tag{2.1}$$

The second identity follows by cancellation at the intermediate vertices, including for backward steps.

For $u \in G$, define

$$s(u) = 1 - \frac{1}{m}\operatorname{Re}\operatorname{tr} u, \qquad \chi(u) = \frac{\|u-1\|_{\mathrm{F}}}{\sqrt{m}}\,. \tag{2.2}$$

Unitarity and the trace identity for the Frobenius norm give

$$\chi(u)^2 = 2s(u), \qquad 0 \leq s(u) \leq 2, \qquad \chi(uv) \leq \chi(u) + \chi(v)\,. \tag{2.3}$$

Indeed, $\|u-1\|_{\mathrm{F}}^2 = 2m - 2\operatorname{Re}\operatorname{tr} u$, and $\|u-1\|_{\mathrm{F}} \leq 2\sqrt{m}$. For the last inequality write $uv - 1 = (u-1) + u(v-1)$ and use invariance of the Frobenius norm under unitary multiplication. Both s and χ are continuous and invariant under conjugation and inversion.

Let dh be probability Haar measure on G, and let $d\eta_\ell$ be its product over the $4\ell^4$ positive links. Write

$$\begin{aligned} U_{y;\mu\nu} &= U(y,\mu)U(y+e_\mu,\nu)U(y+e_\nu,\mu)^{-1}U(y,\nu)^{-1}, \\ E_\ell(U) &= \sum_{y\in\Lambda_\ell}\sum_{\mu<\nu} s(U_{y;\mu\nu}), \\ Z_\ell(\beta) &= \int e^{-\beta E_\ell(U)}\, d\eta_\ell(U)\,. \end{aligned} \tag{2.4}$$

Each site and increasing axis pair contributes once, including when the periodic side is small. There is no further factor in this definition of the energy.

Let H_4 be the group of signed permutation matrices. For $A(y) = hy + z$, with $h \in H_4$ and $z \in \Lambda_\ell$, the pullback A^*U assigns to a positive link its image transport, using the inverse variable if the image direction is negative.

Proposition 2.1 (The lattice law)**.** *For every $\ell \geq 1$ and $\beta \in \mathbb{R}$, the measure*

$$d\mu_{\ell,\beta}(U) = Z_\ell(\beta)^{-1}e^{-\beta E_\ell(U)}\, d\eta_\ell(U) \tag{2.5}$$

is a Borel probability measure on the compact configuration space. Every continuous complex function and every bounded Borel complex function is integrable. Every gauge transformation and every affine signed permutation A preserve this measure. Consequently, for every integrable complex F,

$$\int F(U^\gamma)\, d\mu_{\ell,\beta}(U) = \int F(U)\, d\mu_{\ell,\beta}(U) = \int F(A^*U)\, d\mu_{\ell,\beta}(U)\,. \tag{2.6}$$

Proof. The group G is compact, as a closed subgroup of the compact unitary group. Thus the finite product of link spaces is compact. The energy is continuous and satisfies $0 \leq E_\ell \leq 12\ell^4$, by (2.3) and the six increasing axis pairs. The density is therefore bounded above and bounded below by a positive constant, for every real β. Its integral $Z_\ell(\beta)$ is finite and positive, and division

by that integral proves normalization. Continuous functions on the compact configuration space are bounded; the integrability assertions follow.

Probability Haar measure on a compact group is invariant under right translations and inversion as well as left translations. For example, the image of Haar measure under a fixed right translation is again left invariant with total mass one, so Haar uniqueness gives equality. Inversion then exchanges left and right invariance and also preserves the normalization. Each gauge transformation acts on a link by fixed left and right multiplications. It preserves product Haar measure, while (2.1) makes every plaquette transform by conjugation at its initial vertex. Hence it preserves E_ℓ and the density in (2.5).

An affine signed permutation permutes the positive-link indices and inverts selected link variables. The shift of the initial vertex on a reversed link is included in this permutation. It therefore preserves product Haar measure. The image of an elementary square is another elementary square; changing its initial corner conjugates its transport, and reversing its circulation inverts it. To check the indices, write $he_\mu = \sigma_\mu e_{\pi(\mu)}$, with $\sigma_\mu \in \{-1, 1\}$. The image square has its positive-axis base at

$$hy + z - \mathbf{1}_{\{\sigma_\mu = -1\}} e_{\pi(\mu)} - \mathbf{1}_{\{\sigma_\nu = -1\}} e_{\pi(\nu)}$$

and its axis pair is the increasing rearrangement of $\{\pi(\mu), \pi(\nu)\}$. The axis pairs are permuted, and for each pair the site map is an affine bijection. Thus the face indices are permuted, including on a periodic lattice of side one or two. Conjugation and inversion invariance of s give invariance of E_ℓ. Changing variables in (2.5) proves (2.6). □

2.2 Classical labels

The local polynomial labels describe a connection in radial gauge near one point. They are defined before imposing field equations or taking a continuum limit. For $j \geq 0$, let $\mathcal{A}_j$ be the real vector space of homogeneous polynomial one-forms of degree $j + 1$ with values in $\mathfrak{g}$, satisfying

$$\sum_{\mu=1}^{4} z^\mu A_{j,\mu}(z) = 0 \qquad (z \in \mathbb{R}^4)\,. \tag{2.7}$$

Equivalently, each component is the diagonal evaluation of a symmetric $(j+1)$-linear map, with the same constraint. These are Taylor coefficients, so their definition includes the factorial normalization of a Taylor polynomial. Let $J = (A_j)_{j \geq 0}$ denote a sequence of such coefficients. Choose real coordinates on each $\mathcal{A}_j$, give every coordinate weight $j + 2$, and let V_d be the complex polynomials of total weight at most d invariant under simultaneous conjugation by G.

The ordinary polynomial degree will count factors, separately from weight. An orthogonal matrix acts on one-forms by

$$(T_h A_j)_\mu(z) = \sum_\nu h_{\mu\nu} A_{j,\nu}(h^{-1} z), \qquad (h \cdot v)(J) = v(T_{h^{-1}} J)\,. \tag{2.8}$$

The action preserves the radial condition and commutes with conjugation by G. Complex conjugation of polynomial coefficients is denoted by $v \mapsto \bar{v}$.

Lemma 2.2 (Finite label spaces)**.** *For every $d \geq 0$, the space V_d is finite dimensional and its elements are determined by their values on real jets. It contains* 1 *and is naturally included in V_{d+1}. The decomposition by weight and factor number preserves V_d, and its components are preserved by $O(4)$ and complex conjugation. There exist bases in all degrees, nested as sets of polynomials, consisting of real polynomials homogeneous in both gradings and containing* 1*.*

Proof. Only coordinates with $j + 2 \le d$ can occur in a monomial of weight at most d. There are finitely many such coordinates, and the exponent of each is bounded by d. Thus finitely many monomials span all polynomials under consideration, and their invariant subspace is finite dimensional. A complex polynomial vanishing on every real jet vanishes identically, by the one-variable polynomial identity theorem applied successively in these finitely many real coordinates.

Conjugation by G acts linearly within each $\mathcal{A}_j$. It therefore preserves both weight and factor number, so comparison of homogeneous components of the invariance identity shows that every component remains invariant. The same observation applies to (2.8). All these real linear changes commute with coefficient conjugation. Each invariant homogeneous component is the complex span of its real polynomials, by decomposition into real and imaginary coefficient parts. Choose a real basis in each component, starting with the basis $\{1\}$ of the component with no factors, and retain previous choices as d increases. □

Fix such bases and write $b_{d,i}$ for their elements, with weights $w_{d,i}$ and factor numbers $f_{d,i}$; take $b_{d,0} = 1$. We identify a polynomial with its coefficient vector in this basis when applying linear maps to labels. The finite-dimensional continuum spaces F_d in the theorem will be quotients of labels after a limiting correlation table has been obtained. No such quotient is taken here.

2.3 Local representatives

A classical label is a polynomial on Taylor coefficients; the lattice insertion must instead be a function of finitely many links. We specify the normalization relating the two before constructing such representatives in the next section.

For a finite set $\Sigma \subset \mathbb{Z}^4$, a function is local on Σ if it depends only on positive links whose two endpoints lie in Σ. A polynomial on these links means a polynomial in their matrix entries and conjugate entries. It is continuous, since the coordinate functions are continuous. At $o \in \Lambda_\ell$, place a cover function by substituting $U(o + [x]_\ell, \mu)$ for every cover link (x, μ). This substitution is meaningful even if the stencil wraps around the torus or distinct cover links become the same torus link.

To state the classical normalization, truncate a real jet J to weights at most d and write $A^J_\mu = \sum_{j+2\le d} A_{j,\mu}$. For a spacing parameter $\varepsilon \in \mathbb{R}$, define its link transport by

$$\frac{d}{dt}u(t) = u(t)\varepsilon A^J_\mu(\varepsilon(x + te_\mu)), \qquad u(0) = 1, \qquad U_{J,\varepsilon}(x, \mu) = u(1)\,. \tag{2.9}$$

This convention fixes the order of multiplication in the link transport. For completeness, successive integration gives a normally convergent series on compact parameter sets: the term with r integrations is bounded by $M^r/r!$ if the coefficient matrix is bounded by M on $[0, 1]$. The series solves the equation, and iteration of the integral equation proves uniqueness. In particular, the endpoint is real analytic in ε near 0, for each fixed real jet. Approximating the coefficient by step functions gives convergent products of exponentials from $\mathfrak{g}$; their limits lie in the closed group G. Thus these are actual group-valued transports.

Definition 2.3 (Normalized local section)**.** A normalized local section in degree d consists of a finite H_4-invariant set $\Sigma_d \subset \mathbb{Z}^4$ containing 0 and a complex linear map $v \mapsto P_d[v]$ to polynomials on links with both endpoints in Σ_d. Every $P_d[v]$ is gauge invariant, and

$$P_d[1] = 1, \qquad P_d[\bar v](U) = \overline{P_d[v](U)}, \qquad P_d[v](h^*U) = P_d[h \cdot v](U) \quad (h \in H_4)\,. \tag{2.10}$$

For every real jet J and every $0 \le k \le d$, the section satisfies

$$\frac{d^k}{d\varepsilon^k} P_d[v](U_{J,\varepsilon})\Big|_{\varepsilon=0} = \frac{d^k}{d\varepsilon^k} v(\mathrm{sc}_\varepsilon J)\Big|_{\varepsilon=0}, \qquad (\mathrm{sc}_\varepsilon J)_j = \varepsilon^{j+2} A_j \,. \tag{2.11}$$

The Taylor identity is an equality of classical coefficients at zero spacing, not an equality of observables at nonzero spacing. No compatibility of P_d and P_{d+1} is imposed here. Existence of these sections, with the derivative bounds needed for blocking, is a separate construction. The conditions above alone do not assert it.

2.4 Renormalized insertions

To define a renormalized insertion, one needs an invertible linear map on the entire finite-dimensional label space. It may mix different weights up to d, including the constant label. We specify here how this map enters the lattice observable; its construction and estimates belong to the subsequent ultraviolet sections.

Let $\theta_N > 0$ be the finest-level small-field threshold, and let S_t multiply the basis polynomial $b_{d,i}$ by $t^{f_{d,i}}$. For an invertible complex linear map $\tau_{N,p,d}$ on V_d, define

$$I_{N,p}((d,v),o;U) = a_N^{-d} P_d[S_{\theta_N^{-1}} \tau_{N,p,d}^{-1} v](o;U) \,. \tag{2.12}$$

The map $\tau_{N,p,d}$ reads the running coefficients at a reference level $r(d)$ fixed independently of N, p. More precisely, if $x_m(v)$ are the linear coefficient maps for an insertion transported from level N, their endpoint conditions are

$$x_N(v) = v, \qquad x_{r(d)}(v) = \tau_{N,p,d} v, \qquad x_m(a_N^d 1) = a_m^d 1 \quad (r(d) \le m \le N) \,. \tag{2.13}$$

These conditions concern coefficients of the inserted density, rather than pointwise values of the coarse observable.

Proposition 2.4 (Locality of the insertions)**.** *Given normalized local sections, positive thresholds and invertible maps as in (2.12), the insertions are complex linear in v, continuous, gauge invariant and local on $o + [\Sigma_d]_\ell$, for $\ell = L_p D_N$. They satisfy the literal translation identity*

$$I_{N,p}((d,v),o;U(\cdot + z)) = I_{N,p}((d,v),o+z;U) \,. \tag{2.14}$$

If $N \ge r(d)$, the maps $\tau_{N,p,d}$ commute with conjugation and H_4, and (2.13) holds, then the insertions respect reality and, for $A(o) = ho + z$,

$$I_{N,p}((d,h\cdot v),A(o);U) = I_{N,p}((d,v),o;A^*U), \qquad I_{N,p}((d,a_{r(d)}^d 1),o;U) = 1 \,. \tag{2.15}$$

Their running coefficients at level $r(d)$, with terminal input $\tau_{N,p,d}^{-1} v$, equal v.

Proof. Every assertion in the first sentence follows by composition of the linear coefficient maps with the polynomial section of Definition 2.3. Periodic placement reads only links with both endpoints in the image of Σ_d; identifying distinct cover vertices does not enlarge this set. Translating the placement proves (2.14). Factor scaling commutes with conjugation and with H_4, since these actions preserve factor number. Their commutation with $\tau_{N,p,d}$ and (2.10) give reality and affine covariance. The reference endpoint and unit condition yield $\tau_{N,p,d}(a_N^d 1) = a_{r(d)}^d 1$. Since $S_t 1 = 1$ and $P_d[1] = 1$, substitution in (2.12) proves its unit identity. Finally, $x_{r(d)}(\tau_{N,p,d}^{-1} v) = v$ by (2.13). □

For the token set $\mathcal{T} = \bigsqcup_d V_d$, Proposition 2.4 supplies, under its hypotheses, the continuity, gauge invariance and finite-stencil assertions Reg2–Reg4. They hold at every cutoff; the additional reference normalization is asserted only once $N \geq r(d)$. With integrability supplied by Proposition 2.1, the lattice moments and smeared moments in Part I are therefore ordinary normalized integrals of these polynomial observables.

Unresolved. The existence of normalized local sections with their required derivative bounds, and the same-family construction of the invertible transfer maps satisfying (2.13), remain to be proved in the later ultraviolet sections. The local insertion proposition is conditional on these inputs.[7]

3 Infinite volume

The ultraviolet construction controls local observables independently of the size of the torus. We use this control to choose one volume subsequence for every observable and every separated test. There are two different tasks: obtaining a limit of the correlations, and identifying a derivative of that limit with another field. The first uses reflection positivity and compactness; the second uses equation relations and a finite-dimensional rank comparison.

We first extract an auxiliary spacing subsequence as well as a volume subsequence. Uniqueness of the full fixed-volume limits will remove the spacing subsequence at the end. The Wilson family stays fixed throughout. Only its mixed correlations are constructed here; the phase decomposition requires the rotations and slab operators developed later.

3.1 Lattice estimates

Write V_d for the finite-dimensional coefficient space of degree-d lattice labels. A token is (d, v) with $v \in V_d$, and its insertion $I_{N,p}((d, v), y)$ is linear in v. The construction supplies a distinguished vector $u_d \in V_d$ whose insertion is 1 once N exceeds its reference level $r(d)$. This vector incorporates the reference normalization; it need not be the coefficient vector of the constant polynomial. We suppress the pair (d, v) when its degree is specified. Let H_4 denote the signed permutation matrices, with label action ρ_d, and let a bar denote coefficient conjugation.

The spacings satisfy $a_0 = 1$ and $a_{j+1} = a_j/\ell_j$, with odd integers $\ell_j \geq 3$. The physical periods L_p increase to infinity and are multiples of four. We use gentleness:

$$\frac{\log \ell_j}{\log a_j^{-1}} \longrightarrow 0 \qquad (j \longrightarrow \infty)\,. \tag{3.1}$$

The quotient at $j = 0$ plays no role. The bare couplings are nonnegative. The insertions have fixed finite stencils; their translation, H_4, conjugation and unit identities hold beyond their reference levels. These are properties of the constructed family, beyond public regularity.

For a reflection tiling $\mathcal{C}$ of the fine torus, an interior observable is a bounded measurable gauge-invariant function of links with both endpoints strictly inside its base cell. Reflect it into every cell, conjugating on odd reflections, and denote its copies by F_C. Define

$$\|F\|_{\mathcal{C}} = \left| \left\langle \prod_{C \in \mathcal{C}} F_C \right\rangle_{N,p} \right|^{1/|\mathcal{C}|}. \tag{3.2}$$

[7] Formal correspondence at revision `3456e5e`: Local sections: `YM/Integrated.lean`, lines 214–219. Transfer conditions: `YM/Source/Insertion.lean`, lines 89–114; same-family application: `YM/FullVolume/Cross.lean`, lines 30–71.

We use site and bond seams, with an even number of cells in each direction. The centered tiling $\mathcal{C}_j(y)$ has physical cell side a_j and base cell centered at y. The lattice input is the following estimate. For every $d \in \mathbb{N}$ and $R, \varepsilon > 0$, there exists $A \geq 0$ such that, for every $j \geq r(d)$, eventually in N, uniformly in p, y and $\|v\|_{V_d} \leq R$, the insertion is interior and

$$\|I_{N,p}(v, y)\|_{\mathcal{C}_j(y)} \leq A a_j^{-d-\varepsilon} . \tag{3.3}$$

The cutoff threshold may depend on j. A fixed finite collection of transformed labels is included by enlarging R. [8]

Unresolved. Prove (3.3), gentleness and the stated insertion identities for the same family as Proposition 1.9, using the chosen construction run's centered estimate and insertion construction. The alternative family of unit III.13 does not identify these data.[9]

Lemma 3.1 (Reflection estimates)**.** *For the Wilson law on an even torus and every $\beta \geq 0$, site and bond reflections are positive on interior observables. For every reflection tiling $\mathcal{C}$ and interior base-cell observables F_C,*

$$\left| \left\langle \prod_{C \in \mathcal{C}} (F_C)_C \right\rangle_\beta \right| \leq \prod_{C \in \mathcal{C}} \|F_C\|_{\mathcal{C}} . \tag{3.4}$$

The root in (3.2) is subadditive on these observables.

Proof. The kernel $(U, V) \mapsto \operatorname{Re} \operatorname{tr}(UV^*)$ on G is positive semidefinite: it is the real Euclidean inner product of the matrix entries. Its exponential with a nonnegative coefficient is positive semidefinite by the power series and Schur products. Multiplication by the two half-slice weights preserves positivity. Haar invariance writes the kernel obtained by integrating crossing links as a double average

$$K(X, Y) = \iint K_0(g \cdot X, h \cdot Y)\, dg\, dh .$$

Its quadratic forms are those of K_0 integrated against finite combinations of these averaged point masses, hence are nonnegative. The use of independent variables in the two arguments is essential. Splitting a Wilson configuration into the fixed planes, crossing links and open halves expresses its reflected form as an integral of these positive boundary forms. For a site seam the boundary form is a squared modulus. The change of variables preserves product Haar measure, and division by the positive partition function preserves positivity. All integrands are bounded on compact configuration spaces. [10]

Cauchy–Schwarz bounds the square of the absolute expectation of a mixed cell assignment by the product of its two folded assignments. The finite folding principle on a periodic rectangular tiling says that a nonempty class closed under both folds contains a constant assignment. To see this, fix a coordinate and start with a band of width one cell, on which every assignment is independent of that coordinate. Folding at its initial seam copies its first layer to the preceding layer and enlarges the band by one cell, as long as its width does not exceed half the period. Once the band fills a half-period, one more fold makes the assignment independent of that coordinate everywhere. Each fold preserves independence of the coordinates already treated. Repeat in the

[8] Formal correspondence at revision `3456e5e`: `Extraction/Center.lean`: the uniform centered estimate and its projection to each extraction.

[9] Formal correspondence at revision `3456e5e`: `MergeProof/Gardens.lean` and `Source/Construction.lean`.

[10] Formal correspondence at revision `3456e5e`: `Slice/Positive.lean`: positivity of the weight and double average; `Slice/Reflection.lean`: reflected positivity.

remaining coordinates. Apply this principle to maximizing assignments from any fixed finite list of functions. If their maximum is $M > 0$, the fold inequality $M^2 \le M_1 M_2$, with $0 \le M_i \le M$, keeps both folds maximal; if $M = 0$, every assignment is maximal. Thus a constant assignment attains the maximum. Normalize each F_C by $\|F_C\|_{\mathcal{C}} + \eta$ and let $\eta \downarrow 0$ to obtain (3.4). Finally expand the reflected product of $F + G$, apply (3.4) to every term, and take the $|\mathcal{C}|$th root. [11] □

3.2 Bounds and compactness

Centered estimates control an insertion well inside a translated larger cell. For aligned fine-cell centers, disseminate the larger-cell copies and apply (3.4) on the finer tiling, putting 1 in unused cells. The root is bounded by the largest centered root of the transformed labels. Between consecutive aligned centers, translation invariance and the fold estimate give

$$R(z)^2 \le R(z-1)R(z+1)\,.$$

A nonnegative log-convex sequence is bounded by its endpoint maximum. Applying this in each coordinate proves the translated-cell bound when the anchor is at least $3a_k$ from the boundary and its stencil fits within distance a_k, where a_k is the finer cell side. [12]

Suppose n insertion sites have mutual physical torus sup-distance at least r. Choose $m \le k$ with $a_m < r$ and $6na_k < a_m$. In each coordinate the translations placing an anchor within $3a_k$ of a cell boundary occupy at most $6na_k$ of one period a_m. Choose a translation outside these strips. Different sites occupy different cells, so the translated-cell bound and (3.4) bound their correlation by A^n. Here A bounds the finitely many centered roots at level k, and stencils fit once N is sufficiently large. This gives local bounds at each fixed volume and bounds independent of the volume on every compact subset of Conf_n along ordered tails. An ordered tail means: first the volume index is large, then the spacing index is large with a threshold allowed to depend on that volume.

For compactness we initially restrict to

$$\mathrm{Gen}_n = \{x : x_i^\mu \ne x_j^\mu \text{ for } i \ne j,\ \mu = 1, \dots, 4\}\,.$$

At fixed period replace inequality of coordinates by inequality modulo $L_p\mathbb{Z}$, and denote the set by $\mathrm{Gen}_n^{(p)}$. Their complements are countable unions of hyperplanes and have measure zero.

Equicontinuity on these sets comes from the positive transfer kernel in one coordinate direction. Integrating within slices and assigning half the slice action to each neighbor expresses a correlation of slab observables as

$$\frac{\mathrm{Tr}(B_1 K^{Y_1} \cdots B_n K^{Y_n})}{Z}, \qquad Z = \mathrm{Tr}\, K^M > 0, \qquad nw + \sum_i Y_i = M\,.$$

Here w is the common slab width in lattice steps, Y_i are the intervening gaps, and traces are cyclic kernel integrals. The slice space is compact and the kernels are continuous. For $M = 2qw$, the column estimate is

$$\big(\mathrm{Tr}(B_i^* B_i)^q / Z\big)^{1/(2q)} \le A_i\,.$$

[11] Formal correspondence at revision `3456e5e`: `Cells/Flood.lean`: the folding principle; `Cells/Chess.lean`: maximization, the product estimate and subadditivity.

[12] Formal correspondence at revision `3456e5e`: `Extraction/Around.lean`: transfer to interior positions; `Cells/Slide.lean`: translation log-convexity.

Apply (3.4) to $2q$ alternating copies of the observable and its reflected conjugate. Divisibility of the period by four supplies the integer q when the slabs have twice a centered cell's width.

Transferring $h \le b$ steps between two gaps, with all original gaps greater than $2b$, changes the normalized trace by at most $(\prod_i A_i)h/b$. Here is the estimate that avoids a slice-dimension factor. Replace all cyclic integrals by one common finite quadrature. For its positive matrix K, define $K^z = \exp(z \log K)$ on the positive eigenspaces and zero on the kernel. The trace with two exponents $Y_i + z, Y_j - z$ is entire. For $|z| \le 2b$, Schatten Hölder bounds it by

$$\prod_i \|B_i\|_{2q}\, Z^{1-n/(2q)}, \qquad \frac{n}{2q} + \frac{\sum_i \operatorname{Re} Y_i(z)}{M} = 1\,.$$

Imaginary powers are contractions, and the real powers have the indicated Schatten norms. Cauchy's estimate on discs of radius b centered on $[0, h]$ bounds the derivative by this expression divided by b. Integrate and pass to the common quadrature limit. Every endpoint exponent is positive, so the convention on the kernel changes no endpoint value. [13]

Moving one site changes two gaps. In physical units the preceding estimate gives a Lipschitz bound; telescoping over the $4n$ coordinates gives equicontinuity on generic compact sets. The centered estimates make its constant independent of the volume along ordered tails. For $n = 1$ use translation invariance; for $n = 0$ the moment is 1. No derivative bound has been used.

Proposition 3.2 (Auxiliary extraction)**.** *Assume the lattice specification and (3.3). For every pair of strictly increasing spacing and volume index sequences there exist refinements N_j, p_k, multilinear kernels P_p at every fixed volume, and a multilinear kernel S_0 such that P_p is continuous on $\mathrm{Gen}_n^{(p)}$, the lifted lattice moments at spacing a_{N_j} converge locally uniformly there to P_p, and P_{p_k} converges locally uniformly on Gen_n to S_0. For every token list v and separated test f, the function $fS_0(v;\cdot)$ is integrable, f is eventually p_k-admissible, and numbers $R_p(v;f)$ satisfy*

$$\lim_{j\to\infty} C_{N_j,p}(v;f) = R_p(v;f), \qquad \lim_{k\to\infty} R_{p_k}(v;f) = \int f(x) S_0(v;x)\,dx\,. \tag{3.5}$$

The first limit is required at every admissible fixed volume. One refinement serves every arity, token list and test.

Proof. Extend the lattice functions by rounding points to the grid. Rounding moves a configuration by $O(na_N)$; on a generic compact neighborhood the oscillation of these step functions is therefore bounded by $C|x - y| + o(1)$, uniformly in x, y. Choose a countable dense set in each such neighborhood and extract convergence at its points. A finite net and this oscillation estimate make the subsequence uniformly Cauchy on each compact set; its limit is continuous. This is the compactness argument underlying Arzelà–Ascoli, without requiring the step functions themselves to be continuous. Apply it over all fixed periods, arities, degree lists, basis labels and generic compact exhaustions. These indices are countable, so one spacing subsequence gives all the kernels P_p. The volume bounds and moduli pass to P_p by taking the spacing limit at moving lattice points. A second countable extraction gives the volume subsequence and S_0. Extend from basis labels by multilinearity. Set the raw kernels equal to zero off their generic sets; these measurable extensions change no integral.

Write each weighted lattice sum as an integral of a step function. A compact neighborhood of $\operatorname{supp} f$ remains separated; at an admissible fixed volume it also avoids periodic collisions. Local

[13] Formal correspondence at revision `3456e5e`: `Complex/Gaps.lean`: trace bounds and finite differences; `Slice/Calendar.lean`: kernel passage; `Extraction/Difference.lean`: columns and one-coordinate shifts.

bounds dominate the integrands there, and rounding changes f uniformly by a quantity tending to zero. Almost-everywhere convergence on $\mathrm{Gen}_n^{(p)}$ gives the first limit by dominated convergence. Apply dominated convergence again on $\operatorname{supp} f$, using the volume-independent bound and the full-measure set Gen_n. This gives the second limit and integrability. Eventual admissibility follows from $L_{p_k} \to \infty$. [14] □

3.3 Equation relations

Compactness does not yet identify derivatives at configurations where coordinate projections coincide. A raised-degree insertion represents the derivative, but its coefficient may grow with the cutoff. Equation relations remove the invisible part; a separate rank bound controls what remains. We specify the additional input.

Along the refined ordered tails let idempotents $P_{d,N,p} : V_d \to V_d$ converge in operator norm to an idempotent P_d. A coefficient family $w_{N,p} \in V_d$ is admissible for this comparison if, for every $\varepsilon > 0$, eventually

$$\|B_{d,N,p}^{-1} w_{N,p}\|_{V_d} \le a_N^{-\varepsilon} .$$

Here $B_{d,N,p}$ is the terminal transfer from cutoff to reference coefficients. Its column estimates imply this condition for bounded coefficient families: the logarithm of the product of column bounds is $o(\log a_N^{-1})$.

Call an error $E_{N,p}(y)$ small in reflected cells if one nonnegative rate $\eta_{N,p} \to 0$ along ordered tails has the following property. For every integer $q \ge 0$ and lower level m, there exists $k \ge m$ such that eventually, for every translated bond cell of side a_k and every anchor with relative lattice coordinates z satisfying

$$M \le z^\mu, \quad z^\mu + M + 1 \le a_k/a_N, \quad M = \left\lfloor \frac{a_k/a_N}{8(q+1)} \right\rfloor ,$$

the error is interior and its root is at most $\eta_{N,p}$. At every degree we require:

(i) Each fixed degree-d insertion has an admissible degree-e representative for $d \le e$, differing by an error small in reflected cells.

(ii) The forward difference $a_N^{-1}(I(v, y + e_\mu) - I(v, y))$ has such a representative of degree $d + 1$.

(iii) An admissible family in $\ker P_{d,N,p}$ differs by such an error from bounded measurable observables whose correlations with every fixed spectator list vanish exactly, eventually at each positive separation.

Boundedness in (iii) is only at each cutoff and site, which suffices for integration against the finite Wilson law.

For every one-slot separated smear let $L_{N,p} : V_d \to \mathbb{C}$ be its coefficient map and L its limit computed with S_0. Set $N_d = \bigcap_L \ker L$, including every arity, slot, spectator list and separated test. The additional rank input is

$$\dim_{\mathbb{C}} N_d \le \dim_{\mathbb{C}} \ker P_d . \tag{3.6}$$

The projectors and this inequality are obtained on a further common refinement, retaining all previous limits.

[14] Formal correspondence at revision `3456e5e`: `Extraction/Lipschitz.lean`: the two extractions; `Extraction/Integrate.lean`: dominated convergence; `Extraction/Held.lean`: their combination.

Unresolved. Supply comparisons (i)–(iii), the terminal column bound, and the finite detector argument for (3.6) for the chosen family. The dimension inequality is additional to the centered-root estimate.[15]

Lemma 3.3 (Fixed representatives)**.** *Under these additional inputs, $N_d = \ker P_d$. There exist complex-linear maps $\kappa_{d,e} : V_d \to V_e$ for $d \le e$ and $\tau_{d,\mu} : V_d \to V_{d+1}$ such that every separated one-slot smear satisfies*

$$\int f S_0(v_{i\to\kappa v_i}) = \int f S_0(v), \qquad \int f S_0(v_{i\to\tau_\mu v_i}) = -\int (\partial_{i,\mu} f) S_0(v)\,. \tag{3.7}$$

Here $v_{i\to w}$ means replacement of slot i by w.

Proof. An error small in reflected cells has vanishing separated smearings. Choose its cell below the fixed separation and a finer centered scale for the spectators. Avoid boundary strips as before, with margin M; the factor $8(q+1)$ permits this for any fixed number of slots. The translated-cell estimate and (3.4) bound the mixed moment by $C\eta_{N,p}$. The weighted count of sites meeting $\operatorname{supp} f$ is bounded by the volume of a fixed neighborhood, independently of N, p. Thus its smear tends to zero.

The maps $L_{N,p}$ converge in operator norm because they converge on a finite basis. For fixed v, the bounded family $v - P_{d,N,p}v$ belongs to the projection kernel. Comparison (iii) gives $L(v - P_d v) = 0$ for every L. Thus $\ker P_d \subset N_d$, and (3.6) gives equality.

Summation by parts on the torus gives

$$a_N^{4n} \sum_y f(\widehat{y}) \Delta_{i,\mu} M_{N,p}(v; y) = a_N^{4n} \sum_y \frac{f(\widehat{y} - a_N e_{i,\mu}) - f(\widehat{y})}{a_N} M_{N,p}(v; y)\,.$$

Here $\Delta_{i,\mu}$ is the forward quotient in slot i, and $e_{i,\mu}$ its unit coordinate vector. The test support stays inside the fundamental box eventually, so there is no wrapping. Its backward quotient tends uniformly to $-\partial_{i,\mu} f$ on a fixed neighborhood. Local boundedness and the bounded weighted site count make the error vanish. Comparisons (i)–(ii) therefore give coefficient families $w_{N,p}$ with the required target limits for every spectator smear at once.

Write $p_{N,p} = P_{e,N,p} w_{N,p}$, where e is the representing degree. Comparison (iii) applied to the normalized remainder gives

$$L_{N,p}\left(\frac{w_{N,p} - p_{N,p}}{1 + \|p_{N,p}\|_{V_e}}\right) \longrightarrow 0\,.$$

Indeed division preserves admissibility, and the normalized projected vector is bounded. Take a cluster limit (c, z) of

$$\left(\frac{1}{1 + \|p_{N,p}\|_{V_e}}, \frac{p_{N,p}}{1 + \|p_{N,p}\|_{V_e}}\right)$$

in $[0,1] \times \overline{B}_{V_e}(0,1)$, using a refining ultrafilter of the ordered tails. Then $P_e z = z$, $c + \|z\|_{V_e} = 1$, and $Lz = c\, t_L$ for every target t_L. If $c = 0$, every L annihilates z, so $z \in N_e = \ker P_e$ and $z = 0$, a contradiction. Thus z/c represents every target simultaneously. Choose representatives on each source basis and extend linearly. No extraction over the uncountable collection of spectator tests is needed. [16] □

[15] Formal correspondence at revision `3456e5e`: `MergeProof/Powder.lean` and `Next/RANK/Beam.lean`: same-family assembly and retained rank comparison.

[16] Formal correspondence at revision `3456e5e`: `Work/D01.lean`–`D04.lean`: coefficient costs, errors and differences; `Dash/Limits.lean`: normalized compactness; `Dash/Inert.lean`: fixed representatives.

3.4 Smooth fields and positivity

Proposition 3.4 (Smooth mixed hierarchy)**.** *Under the preceding inputs there exists a multilinear table S, smooth on every Conf_n, equal to S_0 on Gen_n, with the same ordered smeared limits. Its null spaces are N_d. The quotient fields $F_d = V_d/N_d$ have compatible injective inclusions and linear derivatives with*

$$S(\dots,\kappa_{d,e}v,\dots;x) = S(\dots,v,\dots;x), \qquad S(\dots,\partial_\mu A,\dots;x) = \partial_{i,\mu} S(\dots,A,\dots;x)\,. \tag{3.8}$$

The first identity uses coefficient representatives; the second uses quotient fields. The table is normalized, unital, permutation symmetric, translation invariant, H_4 covariant and compatible with conjugation. It is reflection positive in every coordinate direction, for point words and for joint separated tests supported in the positive half-space. For every arity, fixed label list and positional derivative order, there exist $C \geq 0$ and $b \in \mathbb{N}$ such that its derivative norm is at most Cr^{-b} when mutual sup-distances are at least r, $0 < r \leq 1$. The constants are independent of the positions and r.

Proof. Equation (3.7) represents every weak coordinate derivative by another locally integrable kernel. Iteration gives derivatives of every order. For a smooth cutoff χ supported in Conf_n, each derivative of $\chi S_0(v)$ is a finite sum of integrable compactly supported kernels. Integration by parts in the Fourier transform gives, for every integer $b \geq 0$,

$$\sup_\xi (1+|\xi|)^b \left|\widehat{\chi S_0(v)}(\xi)\right| < \infty\,.$$

Fourier inversion supplies a smooth representative. Choose $\chi = 1$ near an arbitrary distinct-point configuration. The local representatives agree almost everywhere on overlaps, hence everywhere there by continuity. They glue to S and equal S_0 on the generic set. The same argument preserves multilinearity and smearings. The coefficient identities preserve null vectors and descend to the quotients. Their characterization by all spectators makes the inclusions injective and compatible. [17]

Exact lattice symmetries and unit identities pass at generic grid configurations and grid translations. Density and continuity give these identities throughout Conf_n and for every real translation. For reflection positivity, fix finitely many joint tests supported in positive-side separated configurations. Their smeared lattice observables are strictly inside the positive half-torus for large p and then large N, since their stencils shrink physically. Lemma 3.1 makes their Gram matrix positive semidefinite. Every entry is a separated smear: its two halves are internally separated and lie strictly on opposite sides of the plane. Pass each entry to its ordered limit. The limiting integrals converge absolutely by smoothness on their compact supports, and the positive-semidefinite cone is closed. Normalized smooth bumps concentrating at positive words give point reflection positivity.

For the position-independent bound, gentleness implies $a_{j+1} \geq c a_j^2$ for some $c > 0$, after adjusting finitely many levels. The first $j \geq j_0$ with $a_j < \delta \leq 1$ therefore satisfies $a_j \geq c_{j_0}\delta^2$. For fixed n, v set $D = 1 + \max_i d_i$. Choose the first sufficiently large m with $a_m < r/2$, then the first $k \geq m$ with $a_k < a_m/[6(n+1)]$. The two first-crossing bounds give $a_k \geq c' r^4$. Equation (3.3) with $\varepsilon = 1$ and the separated-cell argument yield

$$|S(v;x)| \leq C r^{-4nD} \qquad (0 < r \leq 1,\ \min_{i\neq j} |x_i - x_j|_\infty \geq r)\,.$$

[17] Formal correspondence at revision `3456e5e`: `Dash/Regular.lean`, `Dash/Twist.lean`, and `Org/Input.lean`: smooth representatives with the retained extraction.

First pass the estimate at generic points, then use density, retaining separation $r/2$ before taking limits. Repeated derivative identities apply this bound to finitely many raised-label lists. A finite-dimensional norm comparison then bounds each Fréchet derivative norm. Arity zero uses $S_\varnothing = 1$. [18] □

The all-spectator characterization also gives the field symmetries. Inclusions commute with conjugation and H_4 and preserve unit classes. Derivatives commute with conjugation; the chain rule in $S(\rho(h)v; hx) = S(v; x)$ gives

$$\rho_{d+1}(h)\partial_\mu = \sum_\nu h_{\nu\mu}\partial_\nu \rho_d(h) \qquad (h \in H_4)\,.$$

Continuous rotation covariance will require a further argument. [19]

3.5 Removing the spacing refinement

Use the full fixed-volume convergence of Proposition 1.9. For an admissible test at volume p, put

$$\widehat{R}_p(v; f) = \lim_{N\to\infty} C_{N,p}(v; f)\,.$$

Restriction to N_j preserves this limit. Uniqueness in $\mathbb{C}$ and (3.5) give $\widehat{R}_p(v; f) = R_p(v; f)$ at every admissible volume. The outer limit and smooth table are retained. Start the extraction with the full spacing sequence and the prescribed volume sequence p, and compose all the volume refinements into one strictly increasing q. Then

$$\lim_{k\to\infty} \widehat{R}_{p(q(k))}(v; f) = \int f(x)S(v; x)\,dx\,. \tag{3.9}$$

Numbers at inadmissible volumes may be assigned arbitrarily. This is the ordered Wilson limit with the full inner sequence, for every token list and separated test. [20]

Unresolved. The full-spacing fixed-volume theorem for the chosen run remains to be proved in the ultraviolet part, using its fixed-level Cauchy estimates and positional-limit construction. The uniqueness argument above uses this input; it does not prove it.[21]

The smooth null-space definition agrees with the one using separated tests. Consequently token projection $v \mapsto [v]$ identifies lattice representatives with quotient fields in every separated correlation. Multiply that equality by f to transport (3.9) to the field correlations in Proposition 1.10. Its remaining phase assertions require the later construction on this same table.

References

[1] OpenAI, *Lean construction of four-dimensional Yang–Mills theory*, Source release, revision `3456e5e`, 2026, https://github.com/yang-mills-group/OpenAI-YM-20261028.

[18] Formal correspondence at revision `3456e5e`: `Extraction/Harvest.lean`: symmetry and positivity; `Work/D05.lean`: two-scale bound; `Work/B10.lean`: derivative bounds.

[19] Formal correspondence at revision `3456e5e`: `Work/D06.lean`: orientation of the derivative maps.

[20] Formal correspondence at revision `3456e5e`: `FullVolume/Finale/Program.lean`: removal of the spacing refinement; `HierarchyBridge.lean`: public ordered Wilson limit.

[21] Formal correspondence at revision `3456e5e`: `FullVolume/Fields/Filled.lean`: full-spacing fixed-volume convergence for the chosen run.